KPS PUBLIC SCHOOLS • STUDENT TECH BENCH PROGRAM

Lenovo 500e Chromebook

Solder-Free Repair Guide



Every field-replaceable repair that needs a screwdriver, not a soldering iron.
Lenovo 500e Chromebook, first generation (type 81ES): convertible with stylus.

In this guide

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Built from Lenovo's Hardware Maintenance Manual (500e Chromebook, January 2018).
Part numbers: look up by serial on pcsupport.lenovo.com before ordering.

1 Read first

This guide covers every Lenovo 500e Chromebook repair the Tech Bench can do with hand tools: swapping a part, reseating a cable, or fixing software. Anything that needs a soldering iron, hot air, or board-level work is in Section 6 and goes to the lead tech or the manufacturer.

How this guide was built. Screw counts, sizes and the order of removal come from Lenovo's Hardware Maintenance Manual for the 500e Chromebook (type 81ES). This guide rewrites Lenovo's procedures step by step for a student bench and adds tests and checks. Where Lenovo's manual doesn't give a count, the step says so and the first-teardown notes box is there to record it.

Don't use the Developer Mode battery steps. Lenovo's manual offers ways to switch the battery off in software. On district devices, unplug the battery connector instead: never enter Developer Mode.

Bench levels

Level	Who may do it	Typical repairs
LEVEL 1	Any trained student	Software fixes, single keys, bottom cover
LEVEL 2	Student with supervisor sign-off on this repair	Battery, Wi-Fi, touchpad, speakers, I/O board, display parts, hinges
LEVEL 3	Lead tech or adult supervisor only	Heat sink, upper case (keyboard) transplant, liquid damage, system board

Before you touch any device

1. ☐ Check the asset tag and look the device up in the district inventory. Note any open ticket.
2. ☐ Ask the lead tech whether the unit is still under Lenovo warranty or an accidental-damage plan. A warranty claim may cost less than a part, and opening a unit can affect warranty coverage.
3. ☐ Photograph all four sides of the device before opening it.
4. ☐ Record the symptom in the user's own words on the ticket.
5. ☐ Warn the user that a reset erases local files (Downloads). Cloud files in Drive are safe.

Safety rules

- **Power off, unplug, remove any SD card and cables,** then disconnect the battery before any other work inside.
- **Lithium batteries:** never puncture, bend, or pry against the battery. A swollen, hot, hissing, or smoking battery is a Level 3 stop. Move it to the metal battery bin or sand and get an adult.
- **ESD:** wear a grounded wrist strap on an ESD mat. Touch boards only by the edges.
- **Ribbon cables:** unlock every ZIF latch before sliding a ribbon out. A locked ribbon tears.
- **Glass:** wear safety glasses when removing a cracked screen.
- **Before power-on:** check for loose screws inside. Shake gently and listen. A loose screw can short the board.

Tools and supplies

Item	Use	Notes
Phillips #0 and #1	All screws	Lenovo lists PH0/PH1
Plastic pry tool	Cover latches, bezel	Never metal
Plastic spudger, opening picks, plastic tweezers	Clips, latches, ribbons	Only plastic touches connectors
ESD mat + wrist strap	Static protection	
Magnetic screw mat	Keeping screws in order	Screws are often different lengths
90%+ isopropyl alcohol + soft brush	Cleaning contacts, spill residue	
Kapton tape	Re-taping cables and connectors	
Recovery USB or SD card, 8 GB+	ChromeOS recovery	Chromebook Recovery Utility
Known-good USB-C charger	Testing power and charging	
Safety glasses	Cracked-screen removal	
Fire-safe battery bin or sand bucket	Damaged batteries	Required at every bench

2 Know the machine

Specs that matter for repair

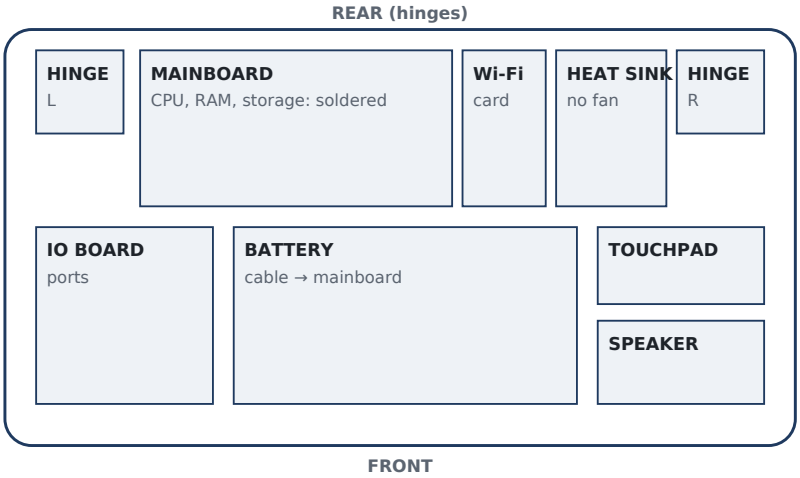
Item	500e	Why it matters
Processor	Intel Celeron N3450	Soldered
Memory / storage	4 GB RAM; 32 GB eMMC	Soldered
Display	11.6" HD IPS touch, 360° hinges	Stylus garage
Access	Bottom cover: 10 × M2.5×6	
Cameras	Front camera + world-facing camera	World-facing camera on the system board side
Wireless	Socketed Wi-Fi card, 1 screw	Replaceable

How it comes apart

A screwed-on bottom cover comes off first. Under it: battery, system board, heat sink, I/O board, speakers and touchpad. The keyboard is built into the upper case (palm rest), the last part left after everything else is removed.

Inside the base

With the cover off you see the parts below. It's a rough map; correct it on your first teardown.



ILLUSTRATIVE — block positions are approximate. Confirm on your first teardown.

500e parts list

Lenovo part numbers vary by configuration. Look up the exact part by serial number before ordering.

Part	Part number	Repair

Look up by serial. Lenovo's FRU numbers depend on the build. Use PartsLookup with the serial on the bottom label.

3 Triage: symptom → repair

Start with the cheapest, least invasive fix. Work through "try first" before ordering parts.

Symptom	Try first	Then	Part likely needed
Won't turn on, no lights	Known-good charger 30 min → S1	R4 → R3	Battery, or system board
Not charging	Other USB-C port and charger → S1	R3 → Sec. 6 if a port is dead	Battery or system board
Battery drains fast / low health in S2	S2 battery test	R3	Battery
Swollen battery / bulging case	STOP — Level 3	R3 by lead tech	Battery
Black screen, power light on	External display via USB-C; S1	R4 → R13 → R12	Display cable or panel
Flicker or lines that change with lid angle	S1	R13	Display cable
Cracked or blotchy screen	—	R12	LCD panel
Bezel loose or broken	Reseat clips	R11	Bezel
Lid floppy, hinge cracked	Check hinge screws	R15	Hinge(s)
One key missing or broken	—	R1	Keycap
Several keys or a row dead	S2 keyboard test; S1; S4	R4 → R16	Keyboard / top cover
Keys type wrong symbols	S4	S4	None
Touchpad dead or erratic	S1; check touchpad settings	R4 → R6	Touchpad or its cable
USB-A / SD card / audio dead	Try another device in the port; S1	R8	I/O board or its cable
Camera not detected	Privacy shutter open? S1	R14	Camera module
No sound (headphones work)	Volume, output; S1; S3	R7	Speakers
Wi-Fi missing or weak	S1 → S3	R5	Wi-Fi card or antennas
Runs hot, fan loud, shuts down	Check vents for lint; S2	R9	Fan or heat sink
Liquid spill	Power off NOW, unplug	R17	Depends on damage
"ChromeOS is missing or damaged"	S3 recovery	R18	System board (storage)
Cracked bottom or lid shell	—	R2	Cover

4 Software repairs

No tools needed. Always do S1 before opening a device.

S1 Hard reset (EC reset) LEVEL 1

Fixes	Won't power on, won't charge, frozen, black screen, unresponsive keyboard or touchpad
Time	2 minutes
Tools	Known-good USB-C charger
Data loss	None

A hard reset restarts the embedded controller (power, keyboard and charging chip). It erases nothing.

Procedure

1. ☐ Unplug the charger and any USB devices. Remove any SD card.
2. ☐ Hold Power for 10 seconds so the device is fully off.
3. ☐ Hold **Refresh** (circular arrow), then tap **Power**. Release Refresh when it starts.
4. ☐ If nothing happens: plug in a known-good charger, wait 30 minutes, and repeat with the charger connected. Try the other USB-C port too.

Verify before closing the ticket

- Device reaches the login screen.
- Charging light comes on with the charger connected.

Tip. After a battery has been disconnected, plug in a charger before deciding the device is dead. Many Chromebooks won't wake on the power button alone.

S2 Built-in hardware diagnostics LEVEL 1

Fixes	Confirms the fault before ordering a part: battery, CPU, memory, keyboard, touchpad, Wi-Fi
Time	5–15 minutes
Tools	None
Data loss	None

Procedure

1. ☐ Sign in with a bench test account if you can.
2. ☐ Open the Launcher, search **Diagnostics**, and open it.
3. ☐ **System:** read battery health and cycle count, then run the battery tests.
4. ☐ Run the CPU and memory tests.
5. ☐ **Connectivity:** confirm Wi-Fi is detected and run the tests.
6. ☐ **Input:** open the keyboard tester and press every key; test the touchpad and touchscreen.
7. ☐ Screenshot any failed result (Ctrl + Show windows) and attach it to the ticket.

Verify before closing the ticket

- Results recorded on the ticket.
- Parts ordered only for a failed test or a fault you saw yourself.

S3 Powerwash and USB recovery — LEVEL 1

Fixes	Slow, crashing, corrupted OS, "ChromeOS is missing or damaged", Wi-Fi missing after an update
Time	Powerwash 10 min; recovery 20–40 min
Tools	Recovery USB or SD card (8 GB+)
Data loss	**Yes.** Erases all local files (Downloads).

District-managed devices. Enrolled devices should force re-enrollment on their first Wi-Fi connection after a wipe. Confirm with the lead tech that forced re-enrollment is on and the bench Wi-Fi reaches enrollment. Never enter Developer Mode on a district device.

Procedure

1. ☐ **Check for updates first:** Settings > About ChromeOS > Check for updates.
2. ☐ **Powerwash:** sign out, press Ctrl + Alt + Shift + R, choose Restart, then Powerwash. (May be blocked by policy; use recovery instead.)
3. ☐ **Make recovery media:** on a working computer, open the Chromebook Recovery Utility, choose the **Lenovo 500e Chromebook** from its list (or search the model), insert an 8 GB+ drive, and click Create now.
4. ☐ Hold **Esc + Refresh**, tap **Power**, and release when the recovery screen appears.
5. ☐ Insert the recovery media and follow the prompts.
6. ☐ After it reboots, connect to Wi-Fi and confirm it re-enrolls to the KPS domain.

Verify before closing the ticket

- Boots to the enrollment or KPS sign-in screen.
- Shows as enrolled in the Admin console.
- Recovery can't write to storage: storage has failed. That's a system board repair or warranty claim.

S4 Keyboard layout and international keyboard fix — LEVEL 1

Fixes	Keys type the wrong symbols (@ and " swapped, accents appear, Z and Y swapped); a replacement keyboard with a different layout
Time	5 minutes
Tools	None
Data loss	None

ChromeOS can have several keyboard layouts installed. Students often switch layouts by accident with **Ctrl + Shift + Space**, or have **US International** on, which turns ' and " into accent keys. The physical keyboard is fine.

Procedure

1. ☐ Look at the bottom-right shelf. A layout code (US, INTL, UK, ES...) means more than one layout is installed.
2. ☐ Open **Settings > Device > Keyboard and inputs > Input** (on older versions: Settings > Advanced > Languages and inputs > Inputs).
3. ☐ Set **English (US)** as the active input. Remove any layouts the student doesn't need, especially US International, unless a teacher asked for it.
4. ☐ Test in a text box: Shift + 2 should type @ and Shift + ' should type ".
5. ☐ If a **replacement keyboard** has a different physical layout (UK keyboard on a US unit, for example), order the correct US part. Changing the software layout won't fix printed key labels.
6. ☐ If the wrong layout keeps coming back for everyone who signs in, it's set by district policy. Tell the lead tech; don't change it on the device.

Verify before closing the ticket

- Every key types its printed character in the S2 keyboard tester and in a text box.
- Only the needed layouts remain installed.

5 Hardware repairs

Most repairs start with the bottom cover (R2) and disconnecting the battery (R3).

Golden rules. Photograph every layer before removing it. Lay screws on the mat in the same pattern as the device. Unlock every connector before pulling a cable. Never force a part: if it won't move, find the screw or clip you missed. Test before the last screws go in.

R1 Single keycap replacement — LEVEL 1

Fixes	Missing, cracked or loose key
Part	Individual key (ask the manufacturer or harvest from a scrapped keyboard)
Time	5 minutes
Tools	Plastic pick, tweezers

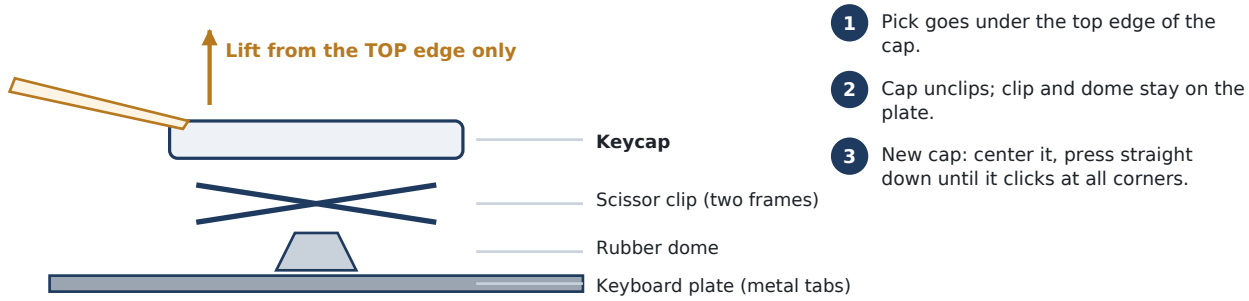


Illustration: side view of one key. Not to scale.

Procedure

1. ☐ Power off. Open the lid flat on the mat.
2. ☐ Slide a plastic pick under the TOP edge of the damaged key and lift until it unclips.
3. ☐ Check the scissor clip and rubber dome under it. A broken clip must be replaced too.
4. ☐ Fit a clip: hook its pins under the metal tabs on the keyboard plate, then press flat.
5. ☐ Center the new key over the clip and press straight down until it clicks at every corner.
6. ☐ Wide keys (Space, Shift, Enter): seat the metal stabilizer wire in its holders first.

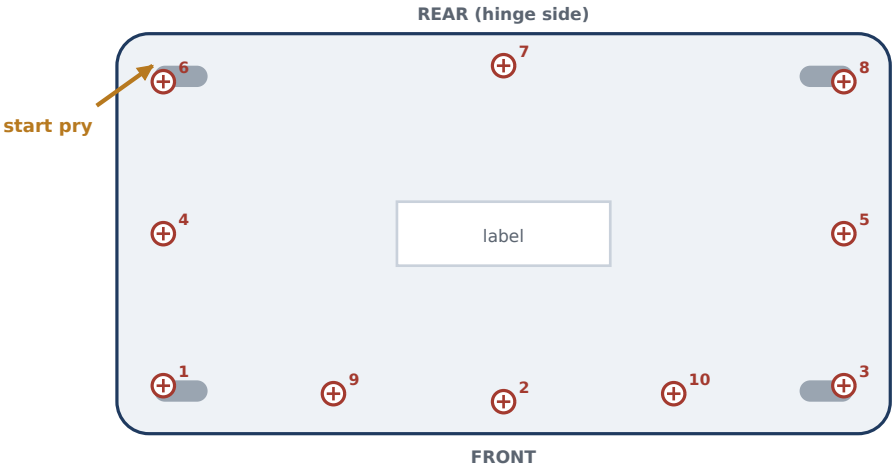
Verify before closing the ticket

- Key clicks evenly and springs back.
- Key registers in the S2 keyboard tester.

R2 Base cover (bottom cover) removal

LEVEL 1

Fixes	Cracked or missing bottom cover; the first step for most internal repairs
Part	Bottom cover (look up by serial)
Time	10 minutes
Tools	Phillips driver, plastic picks, screw mat



1. Remove the 10 bottom screws.
2. Lay screws on the mat in this same pattern.
3. Pry from the corner shown, then work around the edge.
4. Disconnect the battery before anything else.

Illustration: underside. Positions illustrative; count from the service manual.

Procedure

1. ☐ Shut down fully, unplug, and remove any SD card and stylus.
2. ☐ Close the lid and turn the device upside down on the mat.
3. ☐ Remove the 10 M2.5×6 screws.
4. ☐ Release the cover with a plastic pick, starting at a rear corner, then work around the sides. Lift it off.
5. ☐ **Disconnect the battery now** (battery repair, first steps).
6. ☐ To refit: line up the cover, press around the edges until every latch snaps, then refit every screw. Snug only.

Verify before closing the ticket

- No gaps around the seam; device sits flat.
- Every screw refitted, none left over.

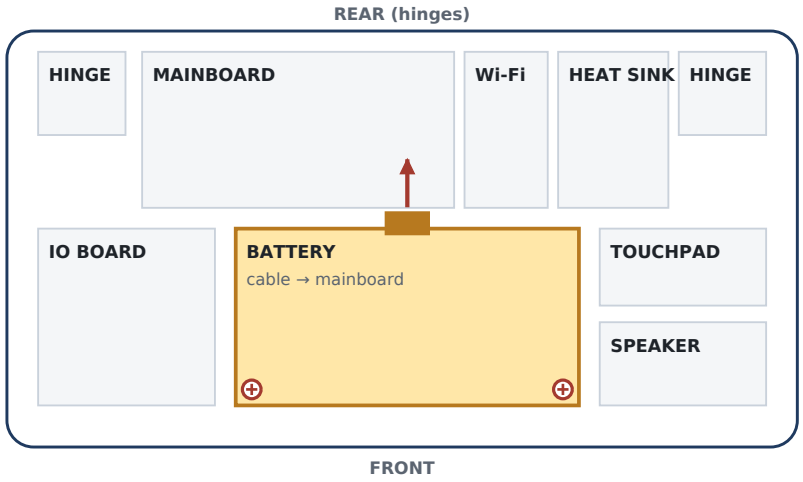
First-teardown notes — fill in the first time you do this repair

Number of screws, and which holes take the longer ones? _____

R3 Battery replacement

LEVEL 2

Fixes	Won't hold charge, low health in S2, swollen battery
Part	Battery (look up by serial)
Time	15–20 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers, ESD strap



1. Unplug the battery cable from the mainboard first (red arrow).
2. Pry at the plug body, never the wires.
3. Remove the battery screws (count them).
4. Lift out flat. Never bend or pry against the cell.

Illustration: base interior, battery highlighted. Screw positions illustrative.

Swollen battery = Level 3. Don't press or pry. The lead tech handles it; the old cell goes straight to the fire-safe bin.

No Developer Mode. Some service manuals disable the battery through Developer Mode or a key sequence. On district devices, unplug the battery connector instead.

Procedure

1. ☐ Remove the bottom cover.
2. ☐ Lift any tape over the battery connector and disconnect the battery cable from the system board.
3. ☐ Remove the 4 M2×4 silver screws.
4. ☐ Lift the battery out. Never bend it.
5. ☐ Fit the new battery, refit the screws, reconnect the cable.
6. ☐ Refit the cover. Plug in the charger before powering on.

Verify before closing the ticket

- Charging light comes on; battery percentage shows.
- S2 battery health near 100% on a new battery.
- Old battery goes to battery recycling, never the trash.

R4 Reseating internal connectors

LEVEL 2

Fixes	Dead keyboard, touchpad or speakers; black screen; dead ports; no power after a drop
Part	None, unless a cable is damaged
Time	15–25 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers, isopropyl alcohol

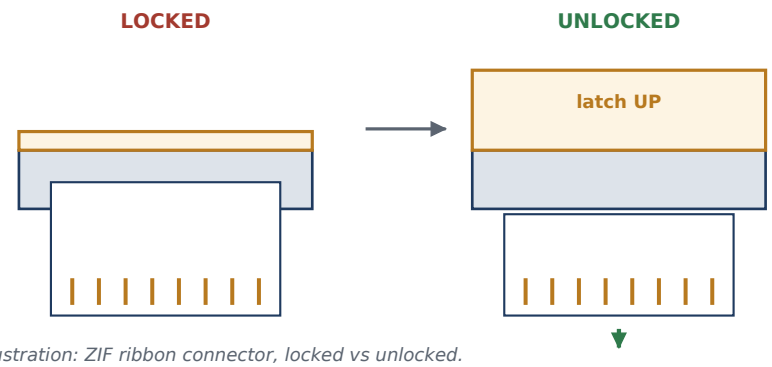


Illustration: ZIF ribbon connector, locked vs unlocked.

1. Flip the hinged latch up (or slide a drawer-type latch out) with a spudger.
2. Ribbon now slides out with zero force. Never pull a locked ribbon.
3. Refit: contacts facing the same way, push in straight and fully, close latch.
4. Board-to-wire plugs (battery, speaker): pry at the plug body, never the wires.

Connector	Type	How to release
Battery cable	Board-to-wire	Lift tape; pry at the connector body
Keyboard ribbon	ZIF ribbon	Unlock the latch; slide out
Touchpad ribbon	ZIF ribbon	Unlock the latch; slide out
Display (eDP) cable	Locking connector	Lift tape and the locking bar, pull straight
I/O board cable(s)	ZIF / FFC	Unlock both ends first
Speaker cable	Board-to-wire	Pry at the plug body
Wi-Fi antennas (2)	Snap-on coax	Lift straight up; white = Main, black = Aux

Procedure

1. ☐ Open the device and disconnect the battery.
2. ☐ Hold Power for 5 seconds with the battery disconnected to drain leftover power.
3. ☐ For each connector related to the symptom: photograph it, unlock and release it, and inspect it for bent pins, burn marks or creases.
4. ☐ Clean dirty contacts with a brush dampened in isopropyl alcohol. Let them dry 5 minutes.
5. ☐ Reconnect each one. Ribbons must go in straight and all the way before you close the latch.
6. ☐ Reconnect the battery last. Plug in and test **before** closing up.

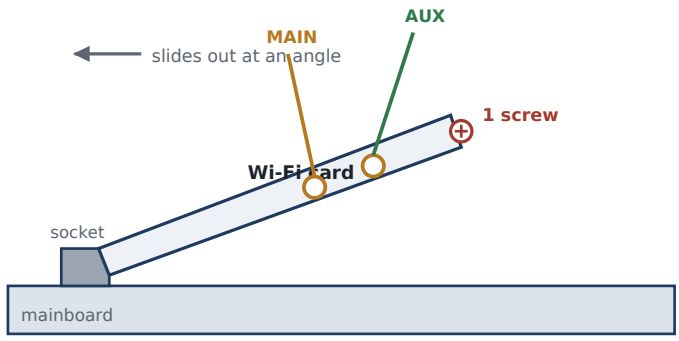
Verify before closing the ticket

- Symptom gone in a full test, not just at boot.
- Creased or torn ribbons replaced.

R5 Wi-Fi card reseat or replacement

LEVEL 2

Fixes	No Wi-Fi or Bluetooth, weak signal, Wi-Fi gone after a drop
Part	Wi-Fi card (look up by serial)
Time	15 minutes
Tools	Phillips driver, plastic tweezers



1. Photograph which antenna goes on which terminal.
2. Lift both antenna plugs straight up with plastic tweezers.
3. Remove the 1 screw. The card springs up at an angle.
4. Slide it out of the socket. Refit in reverse.

Illustration: M.2 Wi-Fi card, side view. Colors of antenna leads vary.

Procedure

1. ☐ Remove the bottom cover and disconnect the battery.
2. ☐ Find the Wi-Fi card.
3. ☐ Note which antenna goes where (white = Main, black = Aux), then lift both straight off.
4. ☐ Remove the 1 M2×3 screw.
5. ☐ The card tilts up. Slide it out of the socket at an angle.
6. ☐ Refit at an angle, press down, refit the screw, snap on both antennas and test.

Verify before closing the ticket

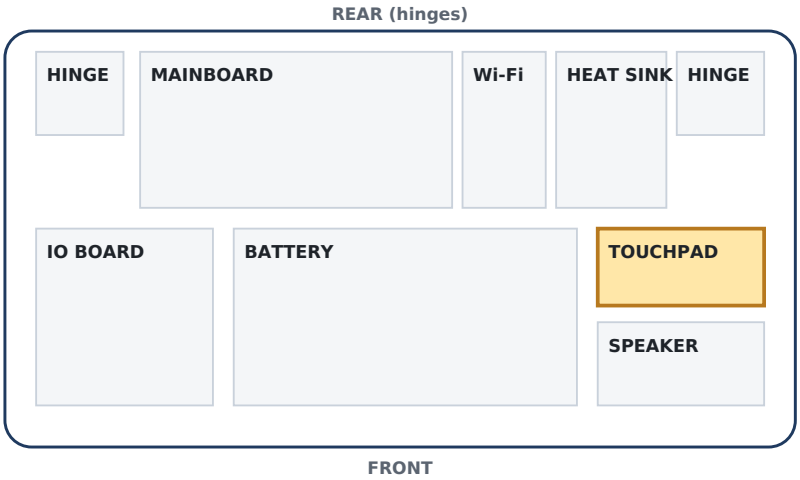
- Wi-Fi connects and the S2 connectivity tests pass.
- Bluetooth finds a device.
- Weak signal after a lid repair usually means a pinched antenna lead.

Approved cards only. Use the Lenovo spare part for this model. Some firmware rejects other cards.

R6 Touchpad and touchpad cable replacement

LEVEL 2

Fixes	Touchpad dead, erratic, cracked or won't click (keyboard works)
Part	Touchpad (look up by serial)
Time	20–30 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers, Kapton tape



1. Battery disconnected first.
2. Unlock the ZIF latch before sliding the ribbon out.
3. Lift tape or Mylar; remove the screws.
4. Conductive tape must touch the pad on reassembly.

Illustration: base interior, touchpad highlighted. Position illustrative.

Check first: touchpad turned on in Settings, S1 done, and the cable reseated.

Procedure

1. ☐ Remove the bottom cover and disconnect the battery.
2. ☐ Unlock the ZIF latch and slide the touchpad ribbon out.
3. ☐ Lift any Mylar or tape, then remove the touchpad bracket screws (count them; record the number in the notes box).
4. ☐ Lift the touchpad out.
5. ☐ Fit the new touchpad, refit the screws and tape, reconnect the ribbon and lock the latch.

Verify before closing the ticket

- Cursor moves, clicks and two-finger scrolls.
- Click works at all four corners.
- Passes the S2 touchpad test.

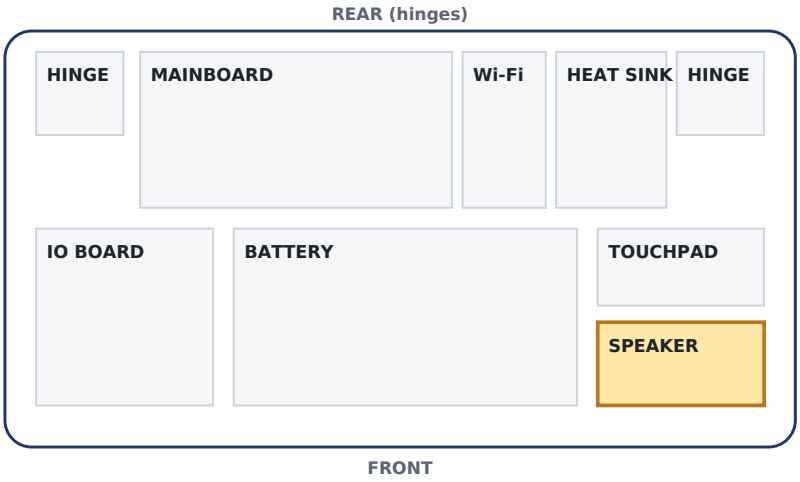
First-teardown notes — fill in the first time you do this repair

Touchpad screw count and any tape positions. _____

R7 Speaker replacement

LEVEL 2

Fixes	No sound or crackling from the speakers, with headphones working
Part	Speakers, left and right pair (look up by serial)
Time	20 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers



1. Battery disconnected.
2. Unplug the speaker cable at the board.
3. Unroute the cable, noting each clip.
4. Keep the rubber isolators: they stop rattle.

Illustration: base interior, speaker highlighted. Position illustrative.

Check first: if headphones also have no sound, the fault is software or the system board, not the speakers.

Procedure

1. ☐ Remove the bottom cover and disconnect the battery.
2. ☐ Disconnect the speaker cable from the system board and free it from its routing clips.
3. ☐ Remove the 4 M2×2.5 screws.
4. ☐ Lift both speakers out together, keeping the rubber isolators.
5. ☐ Fit the new pair, route the cable exactly as before, reconnect.

Verify before closing the ticket

- Clear sound from both sides with no rattle at full volume.
- Headphone jack still works.

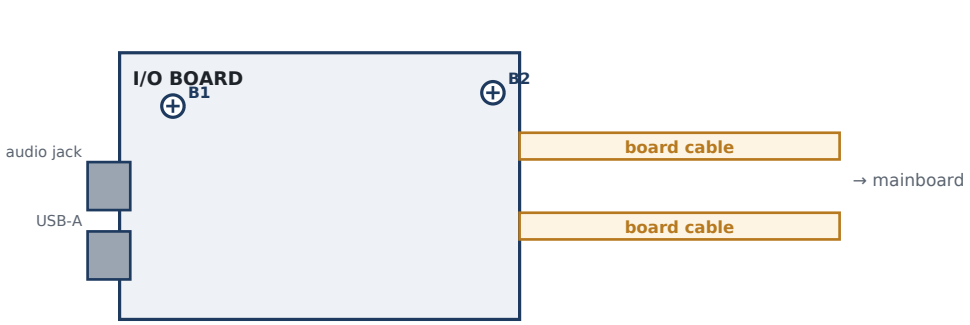
First-teardown notes — fill in the first time you do this repair

Speaker screw count. _____

R8 I/O (USB) daughterboard replacement

LEVEL 2

Fixes	Dead USB-A, audio jack with the system board working
Part	I/O (USB) board and cable (look up by serial)
Time	20–30 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers



1. Remove the board screws (B).
2. Unlock both connectors, then slide out the 30- and 40-pin ribbons.
3. Lift the board, tilting ports out of the case.

Illustration: USB board from above. Port order and screw spots illustrative. Confirm on yours.

Procedure

1. ☐ Remove the bottom cover and disconnect the battery.
2. ☐ Remove the display assembly first (Lenovo's order).
3. ☐ Remove the 4 M2×4 screws.
4. ☐ Disconnect the I/O cable and lift the board out.

Verify before closing the ticket

- Every port on the board works (flash drive, charger, SD card, headphones as fitted).
- Bracket and all screws refitted.

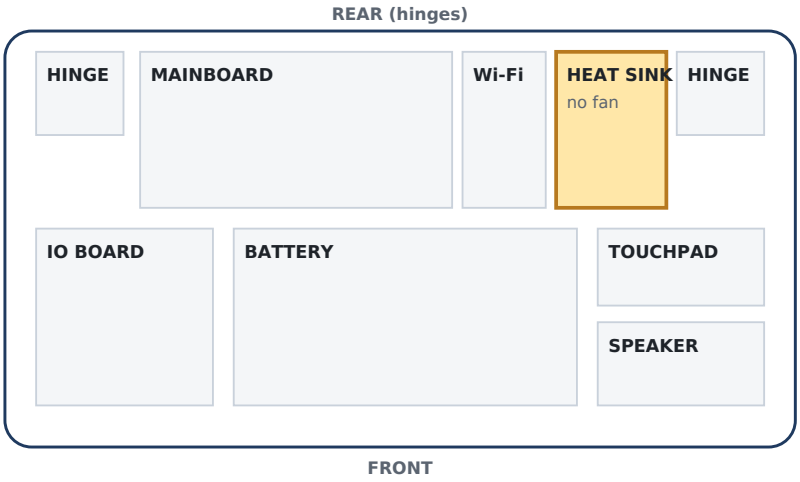
First-teardown notes — fill in the first time you do this repair

I/O board screw count and what ports it carries. _____

R9 Heat sink replacement

LEVEL 3

Fixes	Overheating, thermal shutdowns, loud or rattling fan
Part	Heat sink (look up by serial)
Time	20–30 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers, isopropyl alcohol, lint-free cloth



1. Battery disconnected.
2. Loosen the screws in the numbered order printed on the heat sink.
3. Clean old thermal paste with alcohol.
4. New thermal material, then tighten in order.

Illustration: base interior, cooling highlighted. Position illustrative.

Thermal material. Clean old thermal material off the processor and heat sink every time, and fit new material. Never run the board without the heat sink.

Procedure

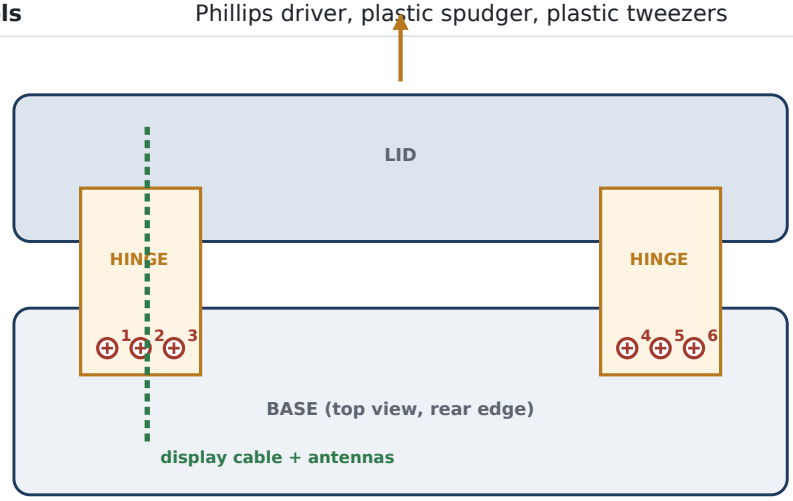
1. ☐ Remove the bottom cover and disconnect the battery. On this model the heat sink comes off with the system board out; follow the board repair, then flip it over.
2. ☐ Loosen the heat sink screws (count them; record the number in the notes box) in the order printed on the heat sink.
3. ☐ Lift off the heat sink. Clean all old thermal material off the processor and heat sink.
4. ☐ Apply new thermal material, then refit, tightening in order.

Verify before closing the ticket

- Device runs a 10-minute video without shutting down.
- Fan (if fitted) spins up under load and is quiet at idle.

R10 Display assembly removal (separating lid from base) LEVEL 2

Fixes	Needed for hinge, lid and some board repairs; replacing a whole display assembly
Part	Display assembly or individual display parts (look up by serial)
Time	20–30 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers



1. Battery out, display cable unplugged at the mainboard.
2. Unthread cable and Wi-Fi antennas from the guides FIRST.
3. Remove the display-to-base hinge screws (count them).
4. Support the lid, then lift it away from the base.

Illustration: hinge area seen from above with the D cover off. Positions approximate.

Procedure

1. ☐ Remove the bottom cover and disconnect the battery.
2. ☐ Disconnect the display cable (lift the tape and locking bar) and both Wi-Fi antennas; free them from their routing.
3. ☐ Open the lid to 90° or more and support it. Remove the 6 M2.5×5 hinge screws.
4. ☐ Lift the display assembly away from the base.
5. ☐ Refit in reverse; route the antennas and display cable back through every clip.

Verify before closing the ticket

- Display, camera and Wi-Fi all work after reassembly.
- Lid opens through its full range without pinching cables.

R11 Display bezel replacement

LEVEL 2

Fixes	Cracked, loose or missing screen bezel; the first step for panel, camera and display-cable repairs
Part	Display bezel (look up by serial)
Time	10 minutes
Tools	Plastic picks, Phillips driver

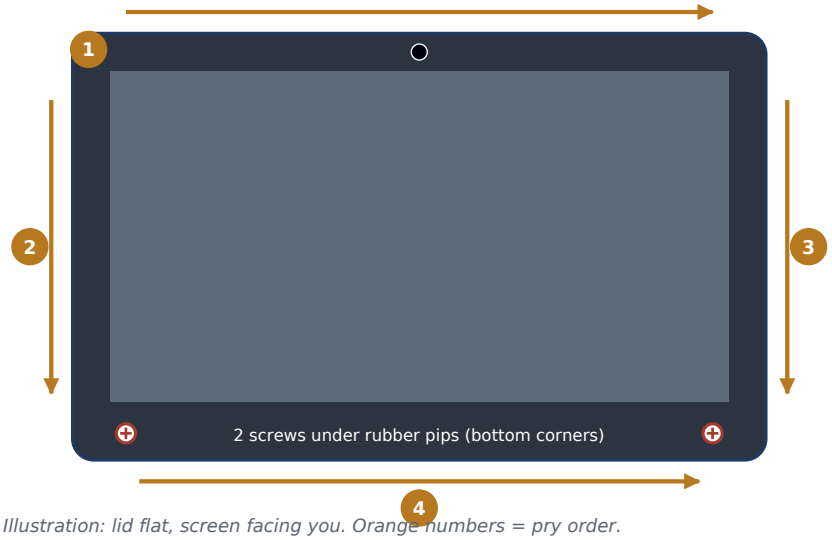


Illustration: lid flat, screen facing you. Orange numbers = pry order.

1. Remove the bezel screws first (under rubber caps if fitted).
2. Start prying at a corner with a flat pick.
3. Work down each side, then across the top.
4. Bottom (hinge) edge last.
5. Don't pull on the camera window.

Procedure

1. ☐ Open the lid flat.
2. ☐ Peel the rubber caps off the strip cover and remove the 2 M2×3 screws under them.
3. ☐ Pry the strip cover and bezel off with a plastic tool.
4. ☐ Refit and press until every clip seats.

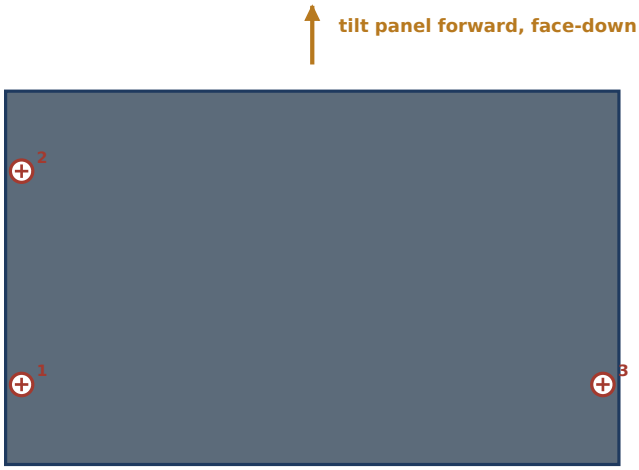
Verify before closing the ticket

- Bezel sits flush all round.
- Camera window lined up; privacy shutter slides freely.
- Lid closes fully.

R12 LCD panel replacement

LEVEL 2

Fixes	Cracked screen, dead pixels, blotches, black screen not fixed by the display cable
Part	LCD panel (match the part number on the back of your panel)
Time	25–35 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers, safety glasses, Kapton tape



1. Remove the 3 panel screws (sides of the panel).
2. Lay the panel face-down on the keyboard (cloth under it).
3. On the back: peel the tape over the connector, open the clip, pull the cable straight out.
4. Connect the new panel and TEST before screwing it down.

Illustration: lid with bezel removed. Screw positions are typical. Count them on yours.

Cracked glass. Wear safety glasses. Tape across a shattered panel before handling it.

Test first: if an external USB-C display works, the system board is fine and the fault is the panel or its cable.

Touch display. On a convertible the touch layer and glass are usually one unit with the panel. Order the touch panel part, not a plain LCD.

Procedure

1. ☐ Remove the bezel.
2. ☐ Remove the 3 (or 4) M2×3 panel screws.
3. ☐ Tip the panel forward and disconnect the LCD cable and the sensor cable.
4. ☐ Connect the new panel and test the picture **before** fixing it down.
5. ☐ Refit everything in reverse.

Verify before closing the ticket

- Full image: no lines, dead pixels or uneven backlight.
- Brightness works across its whole range.
- Image steady through full lid travel.

R13 Display (eDP) cable replacement

LEVEL 2

Fixes	Flicker, lines or black screen that changes with lid angle; external display works
Part	Display (eDP) cable (look up by serial)
Time	40–50 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers, Kapton tape

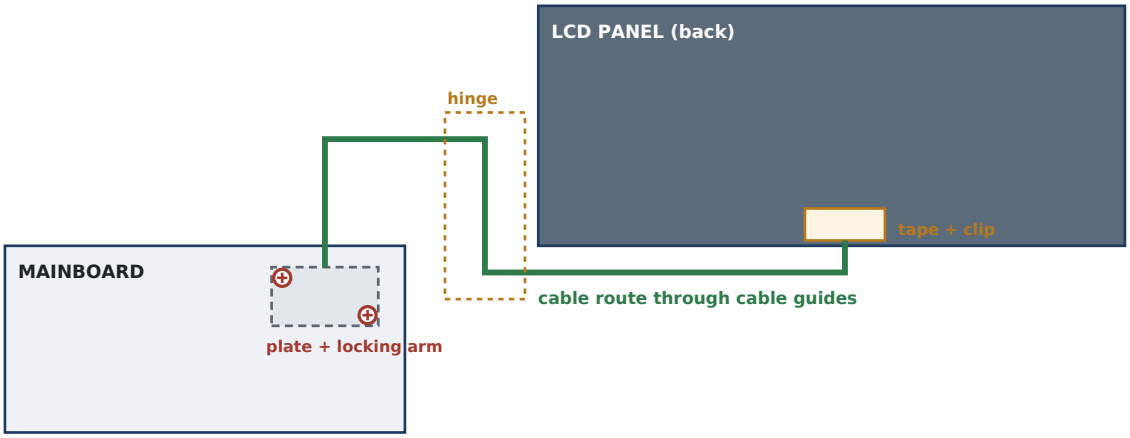


Illustration: display cable path, mainboard to panel (route simplified). Photograph the real route before removing.

Procedure

1. ☐ Remove the bezel and panel; disconnect the display cable at the system board.
2. ☐ Free the cable from the back cover and hinge area; note every clip and tape point.
3. ☐ Route the new cable exactly along the old path and reconnect both ends.
4. ☐ Test at every lid angle before reassembly.

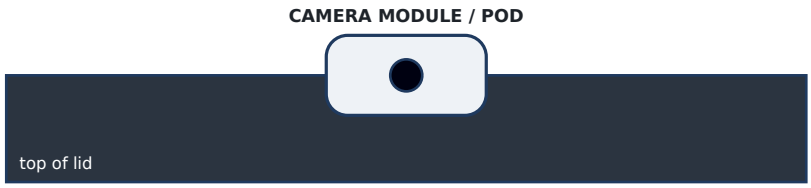
Verify before closing the ticket

- Steady image at every angle: open and close the lid 20 times while watching.
- Camera still works if it shares the cable.

R14 Camera module replacement

LEVEL 2

Fixes	Camera not detected, black image, broken lens
Part	Camera module (look up by serial)
Time	15–20 minutes
Tools	Plastic picks, plastic tweezers



1. Bezel off first (it covers the camera's front).
2. Unplug the small camera cable with tweezers.
3. Lift the module by its board edges, never the lens.
4. Test in the Camera app before closing up.

Illustration: camera at the top-center of the lid. Cap shapes simplified.

Check first: privacy shutter open? Camera works after S1 and a powerwash (S3)?

Procedure

1. ☐ Remove the bezel.
2. ☐ Lift any Mylar or tape over the camera module, disconnect its cable, and lift it off (it's usually held by adhesive).
3. ☐ Fit the new module, reconnect, and test before refitting.

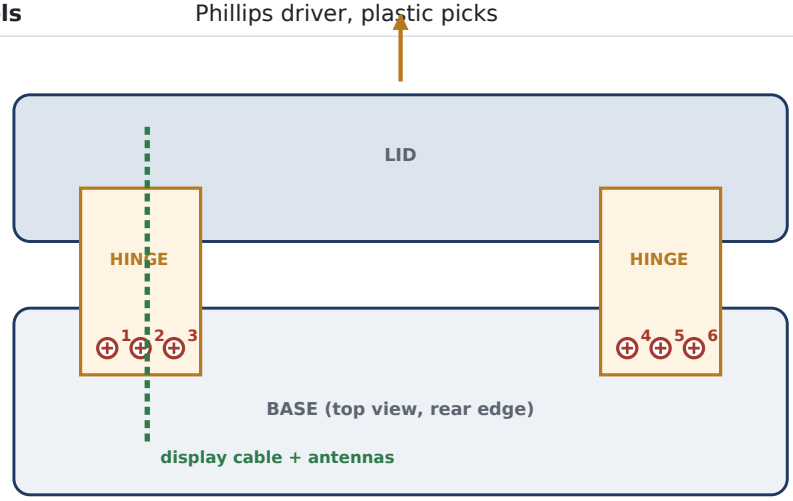
Verify before closing the ticket

- Camera app shows a clear image.
- Privacy shutter fully covers the lens when closed.
- Works in a Google Meet test call.

R15 Hinge replacement

LEVEL 2

Fixes	Floppy lid, cracked hinge, grinding, lid won't stay up
Part	Hinges, left and right (look up by serial)
Time	30–45 minutes
Tools	Phillips driver, plastic picks



1. Battery out, display cable unplugged at the mainboard.
2. Unthread cable and Wi-Fi antennas from the guides FIRST.
3. Base: 6 M2.5×5. Lid side: 6 M2×3 + 2 M2×2.5.
4. Support the lid, then lift it away from the base.

Illustration: hinge area seen from above with the D cover off. Positions approximate.

Check first: tighten the hinge screws. A screw that spins without tightening means a stripped post: that's a cover replacement.

Procedure

1. ☐ Separate the display assembly from the base (display assembly repair).
2. ☐ Remove the 6 M2×3 screws and 2 M2×2.5 screws.
3. ☐ Lift the hinges off the lid.
4. ☐ Fit the new hinges (left and right differ) and reassemble.

Verify before closing the ticket

- Lid holds any angle through its full travel.
- Both hinges feel equally stiff.
- Display steady and Wi-Fi strength normal (no pinched cables).

R16 Keyboard / top cover replacement

LEVEL 3

Fixes	Several dead keys, a dead row, liquid-damaged keyboard, cracked palm rest
Part	Upper case with keyboard (by keyboard language; look up by serial)
Time	60–90 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers, ESD strap, screw mat, Kapton tape

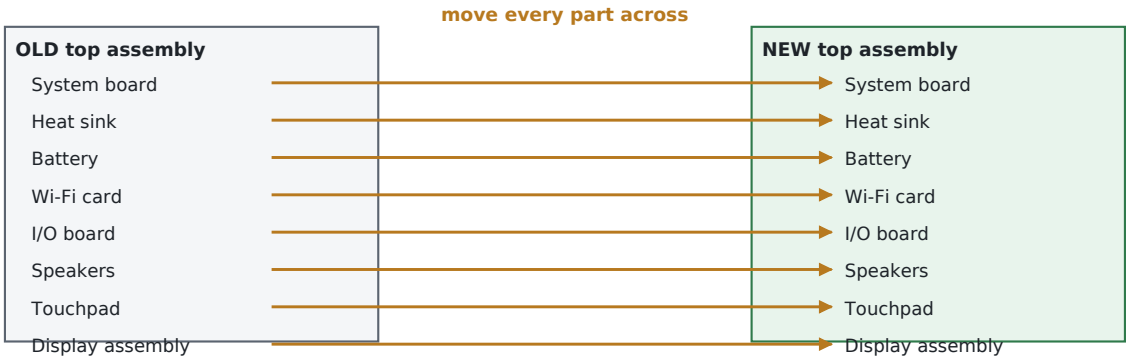


Illustration: what moves to the new top assembly. Check what comes pre-fitted first.

The keyboard is part of the upper case (palm rest). Everything else in the base moves to the new upper case: this is a full transplant.

Procedure

1. ☐ Remove the bottom cover and the battery.
2. ☐ Remove every part listed below from the old upper case, in this order: battery, speakers, Wi-Fi card, display assembly, I/O board, system board (with world-facing camera).
3. ☐ Check what's pre-fitted on the new upper case, then transfer everything else.
4. ☐ Reconnect every cable, checking against your photos.
5. ☐ Reconnect the battery and test everything before closing up.

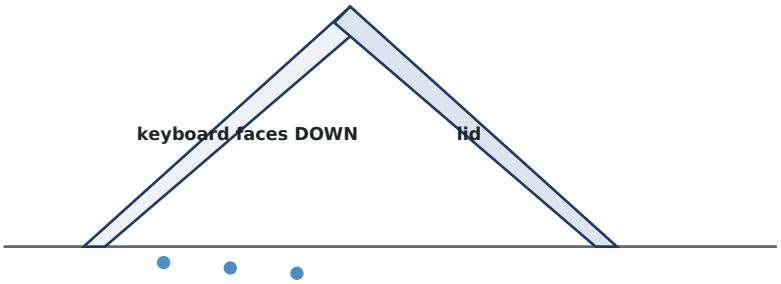
Verify before closing the ticket

- Every key passes the S2 keyboard test.
- Touchpad, display, camera, ports, speakers and Wi-Fi all work.
- No screws left over.

R17 Liquid spill response

LEVEL 3

Fixes	Any liquid on or in the device
Part	Often none; sometimes keyboard/top cover or system board
Time	10 min first response; 24–48 h drying
Tools	90%+ isopropyl alcohol, soft brush, lint-free cloths



1. Power off (hold 10 s), unplug.
2. Tent upside down so liquid drains OUT.
3. Open and disconnect the battery ASAP.
4. Dry 24–48 h. No hair dryer, no rice.

Illustration: drain position after a spill.

Spill-resistant keyboards resist drips, not a full cup. Speed matters more than anything.

Do not power on to test. Current through liquid causes the corrosion that kills boards.

Procedure

1. ☐ Hold Power for 10 seconds to force off. Unplug.
2. ☐ Tent the device upside down (lid at 90°, keyboard down) to drain.
3. ☐ Open it and disconnect the battery as soon as possible.
4. ☐ Blot visible liquid; don't wipe it around.
5. ☐ Sugary or sticky spills: clean affected areas with a soft brush and isopropyl alcohol.
6. ☐ Dry open, battery disconnected, for 24 hours minimum (48 if soaked). No hair dryer, no rice.
7. ☐ Inspect for white or green corrosion; clean connectors.
8. ☐ Reconnect the battery, plug in, and run S2 fully.

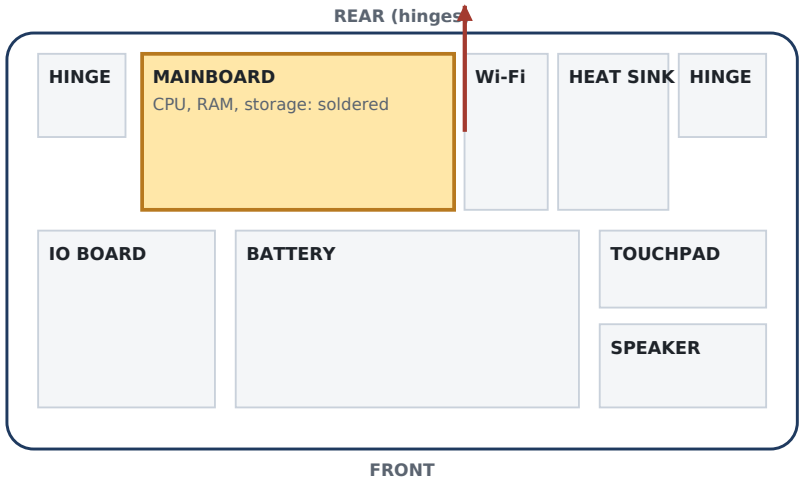
Verify before closing the ticket

- All S2 tests pass.
- Sticky or dead keys: keyboard repair.
- Corrosion on the system board: Section 6.

R18 System board replacement (lead tech)

LEVEL 3

Fixes	Dead USB-C ports, no power with a good battery, failed storage, RAM errors, failed recovery
Part	System board (by configuration; look up by serial)
Time	45–90 minutes
Tools	Phillips driver, plastic spudger, plastic tweezers, ESD strap



1. Battery out first (and the Wi-Fi card if it's socketed).
2. Unplug: display, I/O board, touchpad, keyboard, speaker, antennas
3. Lift by the edges once every screw and cable is free.
4. Lift by the edges onto the ESD mat.

Illustration: base interior, mainboard highlighted. Red arrow = lift side.

Identity and enrollment. Storage, serial number and enrollment live on the system board. The lead tech updates the asset record, deprovisions the old device in the Admin console if needed, and plans re-enrollment. Ask the manufacturer whether the new board ships with the serial set.

Confirm first with S1, S2, S3 and a connector reseat. Most "dead boards" are a battery, cable or connector.

Procedure

1. ☐ Remove the bottom cover, the battery, the Wi-Fi card, the heat sink.
2. ☐ Remove the display assembly and the Wi-Fi card.
3. ☐ Disconnect every cable from the board, including the world-facing camera (photograph first).
4. ☐ Remove the 8 M2×4 screws and lift the board out.
5. ☐ Fit the new board, reconnect every cable, refit the parts removed, then run recovery (S3) if needed and enroll.
6. ☐ Lenovo's HMM has the servicer write the serial and model data (VPD) to a new board with a repair shim. That's a depot step: ask Lenovo or district IT whether the replacement board ships programmed.

Verify before closing the ticket

- Boots, enrolls and passes all of S2.
- Asset record and Admin console show the correct serial number.

6 Out of scope: escalate

These need soldering or board-level work, or parts that aren't sold. Tag the device with the ticket and send it to the lead tech.

Fault	Why	Route
USB-C port loose or dead	Usually soldered to the system board	System board or warranty
Storage or RAM failure	Soldered (system-board level)	System board or warranty
Stylus lost or dead	Not a repair	Order a replacement pen (look up by serial)
Burn marks, melted plastic, burnt smell	Fire hazard	Lead tech immediately
Any repair needing a soldering iron or hot air	Program policy	Lenovo warranty or lead tech

Lenovo support

- **Lenovo support:** pcsupport.lenovo.com (enter the serial number)
- **Parts lookup:** PartsLookup on pcsupport.lenovo.com (by serial)
- **Manual:** the Hardware Maintenance Manual (HMM) for your machine type, on pcsupport.lenovo.com

7 Sources

Checked October 2026. The procedures are the guide author's adaptation of these sources for a student bench; they are not manufacturer-issued instructions.

- Lenovo, *Lenovo 500e Chromebook Hardware Maintenance Manual*, January 2018 (copy at usermanual.wiki) — screw counts and removal order.
- Lenovo PC Support (pcsupport.lenovo.com) — parts lookup by serial.